A few thoughts on Aloni on FC

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1. What's going on with FC in questions?

(1) Are you allowed to eat the apple or the pear?
Yes, I'm allowed to eat the apple, although I'm not allowed to eat the pear.

2. Is wide scope FC really dependent on speaker knowledge?

(2) I'm not sure, but I think (am pretty confident…) that you can go to the beach or you can go to the cinema.

3. Does 'Must P or Q' really entail 'May P and May Q'?

Discourses like 'Must P or Q; but must not P, so must Q' seem totally fine in a way that is very surprising if 'Must P or Q but must not P' is contradictory!

(3) Because you're a resident, you have to file form 1040 or 1040EZ. Moreover, since you have itemized deductions, you can't file 1040EZ.

4. Zimmerman's problem

It seems bad that $\Box\phi \vee_{+} \Box\psi$ stands the same relation to $\Box\phi \wedge \Box\psi$ as $\Diamond\phi \vee_{+} \Diamond\psi$ stands in to $\Diamond\phi \wedge \Diamond\psi$!

Why don't we say 'You must do this or you must do that' to mean what we could also mean by saying 'You must do this and you must do that', that we do say 'You may do this or you may do that' to mean what we could also mean by saying 'You may do this and you may do that'?

5. FC weaker or absent under quantifiers and necessity modals

(4) Every prisoner can use the gym or the exercise yard
(5) Every prisoner can use the gym or can use the exercise yard

This is good news for the idea that FC implications are scalar implicatures, since these work similarly:

(6) Every boy ate some of his lunch

Aloni's system doesn't have quantifiers, but we can see similar effects when we consdi-er disjunctions of possibility modals under a necessity modal, which her system does have:

(7) I must be allowed to use the gym or the exercise yard
(8) I am definitely allowed to use the gym or the exercise yard
(9) You are required to allow me to use the gym or the exercise yard

Let's think through how $\Box(\Diamond\phi\vee_{+}\Diamond\psi)$ works in Aloni's system.

- For s to verify it, each $w \in s$ must be such that every nonempty set of worlds it sees verifies $\Diamond\phi\vee_{+}\Diamond\psi$. In particular, every one-membered set of worlds accessible from it must verify $\Diamond\phi\vee_{+}\Diamond\psi$, which means it must verify $\Diamond\phi\wedge\Diamond\psi$. So $\Box(\Diamond\phi\vee_{+}\Diamond\psi)$ has exactly the same verifiers as $\Box(\Diamond\phi\wedge\Diamond\psi)$, and hence entails it (problematically, given the above).

- For s to falsify it, each $w \in s$ must see some nonempty set of worlds that falsifies $(\Diamond\phi\vee_{+}\Diamond\psi)$, which means it must see a set of worlds that falsifies $(\Diamond\phi\vee\Diamond\psi)$, which means it must see a set of worlds that falsifies both $\Diamond\phi$ and $\Diamond\psi$.

6. Disjunctions under 'must' work weirdly!

The foregoing reasoning about disjunctions of possibility claims under a box carries over to plain disjunctions under a box! s verifies $\Box(\phi\vee_{+}\psi)$ just in case every world in s is such that every nonempty set of worlds it sees verifies $\phi\vee_{+}\psi$. In particular, every one-membered set of worlds it sees must verify $\phi\vee_{+}\psi$, which means these ones must also verify $\phi\wedge\psi$.

- *Upshot:* if ϕ and ψ are incompatible (not both verified by any nonempty state), $\Box(\phi\vee_{+}\psi)$ is inconsistent. But this is bad: 'You must run fast or not run at all' is obviously consistent.
- Also: given the fact below that verifiers are closed under unions, whenever it's true that all the one-membered sets of worlds seen by w verify ϕ, it's also true that all the sets of worlds seen by w verify ϕ. So $\Box(\phi\vee_{+}\psi) \vDash \Box(\phi\wedge\psi)$! ????

7. De Morgan isn't valid

'It's not red or it's not square' isn't a logical consequence of 'It's not red and square'. What should we make of this? Are we just making a mistake when we reason like this?

8. Some properties of the logic [probably already familiar to Maria]

- Every sentence is falsified by the empty state.
- Every sentence without NE is verified by the empty state.
- The union of any number of verifiers is always a verifier; the union of any number of falsifiers is always a falsifier. (Easy induction.)
  - But it's not true that the intersection of two verifiers/falsifiers is always a verifier/falsifier, since the intersection of two nonempty sets can be empty.
  - It's not even true that when two verifiers have a nonempty intersection, this is also a verifier. For example: in a model where a, b, c are each true at just one world, $(a\vee c)\vee_{+}b$ is verified by both $\{a,b\}$ and $\{c,b\}$ but not by $\{b\}$.

- "Convexity" holds: if $s \subseteq t \subseteq u$ and s and u both verify [falsify] ϕ, then t verifies [falsifies] ϕ. Proof:
  - Base case: straightforward for atoms and NE
  - Negation: trivial.
  - Conjunction. Trivial for verifiers. For falsifiers: if s and u both falsify $\phi \wedge \psi$, then there are s' and u' that falsify ϕ, and s'' and u'' that falsify ψ, such that $s=s' \cup s''$ and $u=u' \cup u''$. So by the previous result, $s' \cup u'$ falsifies ϕ, and $s'' \cup u''$ falsifies ψ. So by IH, $t \cap (s' \cup u')$ falsifies ϕ and $t \cap (s'' \cup u'')$ falsifies ψ. But $(t \cap (s' \cup u')) \cup (t \cap (s'' \cup u'')) = t \cap ((s' \cup u' \cup s' \cup u'') = t \cap (s \cup u) = t \cap u = t$. So t falsifies $\phi \wedge \psi$.
  - Disjunction: analogously.
  - $\diamond$: this is a special case of a more general fact, namely that verifiers and falsifiers for $\diamond \phi$ are closed under subsets.
- No nonempty state ever both falsifies and verifies a single sentence. This follows from a stronger fact: no verifier and falsifier of a single sentence ever have a nonempty overlap. Proof:

  Base case: true for atoms and NE

  Conjunction: suppose s verifies $\phi \wedge \psi$ and t falsifies $\phi \wedge \psi$. Then for some u, u', $t=u \cup u'$, u falsifies ϕ, and u' falsifies ψ. But s verifies ϕ, so by IH s does not overlap u, and s verifies ψ, so by IH s does not overlap u'. So s does not overlap t.

  Disjunction: parallel.

  Negation: trivial.

  $\diamond$: suppose s verifies $\diamond \phi$ and t falsifies $\diamond \phi$. Then every world in s sees a nonempty set that verifies ϕ, while every nonempty set seen by any world in t falsifies ϕ. So by IH, no nonempty set is seen both by a world in s and by a world in t. So long as R is serial (*every world sees at least one world*), it follows that s and t do not overlap.
- Corollary: $\phi \wedge \neg \phi \vDash \bot$ for every ϕ. (But it does not follow that $\phi \wedge \neg \phi \vDash \psi$.)